AI-driven land document digitization and marketplace

"im working on the topic of land documents digitization using government server or ocr scanning then validating all the available documents using ai then providing a intelligent marketplace for purchasing using validated documents instead of advertisement modules so can u give the best matching results of research papers matching this field" *Generated by LeapSpace*

AI-Enabled Land Document Digitization and Trusted Property Transactions at a Glance

Key Findings Table

Research cluster	What it contributes to your topic	Best-matching evidence	Relevance to your proposed system
OCR + document understanding for land-related records	Shows that scanning alone is not enough; robust pipelines need preprocessing, OCR, layout parsing, and structured extraction [1] [2] [3]	OCR quality improves with binarization, denoising, skew reduction, and quality control [1] [4]; layout-aware extraction reduces downstream errors [2] [3]	Core for digitizing deeds, titles, registry forms, and scanned records
Map OCR + spatial reasoning	Extracts labels, coordinates, and metadata from maps/plans, which matters for cadastral verification [5] [6] [7]	OCR plus orientation detection and spatial context improves map text extraction [6] [7]	Useful for parcel boundary checking and map-based validation
AI validation before storage/use	Strongest direct match for your “validate with AI first” idea [8]	A 2025 hybrid AI-blockchain land-record framework reports 94.2% OCR accuracy, 90.9% NER F1, 93.4% validation accuracy, and 88.9% anomaly-detection F1 [8]	Best fit for pre-registry validation and trusted listing creation
Fraud/anomaly screening + human review	Supports risk scoring and reviewer prioritization rather than fully autonomous decisions [9] [10] [11]	A municipal property-workflow screening model achieved ROC-AUC 0.778 and flagged 66.8% of cases for prioritized review [11]	Good model for document trust scoring and manual escalation
Record linkage + interoperability	Needed to verify ownership across cadastre, registry, tax, survey, and address systems [12] [13] [14] [15] [16]	LADM, linked data, and schema-integration approaches help unify fragmented land systems [14] [15] [17] [18]	Essential for cross-checking “all available documents”

Research cluster	What it contributes to your topic	Best-matching evidence	Relevance to your proposed system
Digital land administration and verified transaction systems	Shows how validated records can feed citizen-facing land services and transfer workflows [19] [20] [21] [22] [23] [24]	eLAS and blockchain/smart-contract systems improve auditability and process speed, but need legal and institutional support [20] [24] [25]	Closest literature to a trusted property marketplace
Marketplace gap	Literature supports trusted registries and verified transaction platforms more than full AI marketplaces replacing ads [22] [23] [26]	Verified-document transaction flows exist, but end-to-end intelligent marketplace evidence is still limited [22] [23] [26]	Opportunity for new research/product development

Direct Answer

The **best-matching research results** for your topic are not just OCR papers; they are papers on an **end-to-end verified land transaction pipeline**: digitization -> extraction -> validation -> cross-registry checking -> trusted transaction workflow [1] [2] [8] [12].

The **single strongest direct match** is the **2025 hybrid AI-blockchain land-record framework**, because it combines OCR, NER, and anomaly detection for **pre-storage authenticity validation** and reports strong validation metrics: **94.2% OCR accuracy, 90.9% NER F1, 93.4% validation accuracy, and 88.9% anomaly-detection F1** [8]. That is the closest evidence to your idea of validating all land documents with AI before using them in transactions [8].

If you want the **best-matching paper groups** for a literature review or prototype, prioritize them in this order:

1. **AI-validated land-record frameworks** using OCR + NER + anomaly detection before secure storage or transfer [8].
2. **OCR/document-understanding pipelines** for noisy legal-style documents and structured extraction [1] [2] [3] [27].
3. **Record linkage/interoperability papers** connecting cadastre, registry, tax, survey, and address systems [12] [13] [14] [15] [16].
4. **Digital land administration / eLAS papers** on citizen-facing land services and verified digital records [19] [20] [21] [25].
5. **Blockchain/smart-contract transaction papers** for auditability, ownership transfer, and verified property workflows [22] [23] [24] [28].

A strong research framing for your topic is: **"Verified-document-first digital land marketplace: OCR-based digitization, AI authenticity validation, cross-registry consistency checking, and trusted property transaction workflows."** [8] [12]

Confidence level: High - this answer aligns closely with the cross-source synthesis and the most directly relevant land-record validation and transaction studies [8].

Study Scope

- **Time span covered:** mostly 2015-2025, with some foundational record-linkage theory used as background [16] [29] [30].
- **Disciplines covered:** OCR/document AI, fraud detection, land administration, cadastral interoperability, e-governance, blockchain-enabled registries [2] [8] [13] [19] [31].
- **Method types:** system proposals, applied frameworks, reviews, prototype platforms, and some quantitative evaluations with OCR/F1/ROC-AUC/throughput metrics [8] [11] [24].

Assumptions & Limitations

- The evidence is **strong at the system-architecture level**, but **weak on property-specific OCR benchmarks** for deeds, title certificates, and cadastral forms with field-level precision/recall/F1 [32] [33].
- There is **limited direct evidence** for fully realized **AI-driven property marketplaces** that completely replace advertisement-style listings with validated-document-first purchase flows [22] [23] [26].
- Security/compliance evidence is thinner than registry/process evidence; retrieved papers discuss governance and data quality more than zero-trust access control or audit design for automated land-document systems [34] [35].

Suggested Further Research

- Build benchmark datasets for deeds, title certificates, mutation records, encumbrance certificates, cadastral forms, and survey maps with field annotations and fraud labels [32] [33].
- Compare **OCR-only vs OCR+NLP vs multimodal fraud detection** for seals, signatures, altered dates, and forged ownership marks [36] [37] [38].
- Study **cross-registry entity resolution** for multilingual names, missing identifiers, and inheritance chains in land ownership [16] [39] [40].
- Evaluate buyer-facing **trust-based ranking/search** for verified-document property marketplaces [22] [26].
- Report public-sector outcomes such as fraud reduction, transfer-time reduction, reviewer workload, and citizen trust, not just model accuracy [24] [25].

Full Report

Foundational Context

Your proposed system has four linked technical layers:

1. **Digitization** of land documents from scans or government repositories [1] [41].
2. **Document understanding** to turn images into structured fields such as owner names, parcel identifiers, dates, and legal references [2] [3] [27].
3. **Validation** to check authenticity, consistency, and anomalies before storage or transaction use [8].
4. **Trusted transaction workflows** where only validated records can power property discovery, due diligence, and transfer [22] [23] [24].

The literature strongly suggests that **OCR alone is insufficient** because errors from noisy scans, complex layouts, and fragmented text propagate into later decisions unless corrected by layout-aware extraction, rule checks, and cross-registry validation [1] [2] [3] [8]. Taken together, these findings suggest that your idea is best treated as a **pipeline architecture**, not as a single model problem [1] [8].

Confidence level: High - the pipeline pattern is one of the clearest recurring findings across the source set [1] [2] [3] [8].

1. OCR and Document Understanding for Land Records

1.1 OCR pipelines for noisy records

The document-AI evidence shows that OCR quality depends heavily on **preprocessing**, especially for degraded or poorly scanned material [1] [4]. Reported helpful steps include **binarization, denoising, grayscale conversion, thresholding, and skew reduction** [42] [43] [44] [45] [46]. This matters for land records because registry archives are often historical, unevenly formatted, or scanned from paper originals [1] [4].

However, the evidence gap is important: the retrieved OCR studies are mostly on **identity cards, invoices, patient reports, historical pages, and scene text**, not actual deeds or title certificates [43] [46] [47] [48] [49] [50]. So the methods are transferable, but the field-specific benchmark evidence is still missing [32] [33].

1.2 Layout-aware extraction

The stronger pattern is that **OCR should be combined with document understanding** for structured extraction [2] [3]. One example replaced a simple PyTesseract pipeline with **Donut + BERT-NER**, improving accuracy and processing efficiency for invoice-style structured fields [27]. Other studies note that layout-aware models can reduce multi-stage error propagation when handling forms and key-value documents [2] [3].

The evidence also shows a key tradeoff: **LayoutLM-style systems depend on OCR quality**, while Donut-style approaches reduce that dependency, though the retrieved studies do not evaluate them directly on property records [33] [51] [52]. One invoice-domain result reported **LayoutLMv3 precision 0.948 and recall 0.611**, with rotation and structural variability as bottlenecks [53]. That is useful as a cautionary signal for land records, which also have structural variation [53].

1.3 Post-processing and structured outputs

OCR output is often noisy and fragmented, and one pipeline improved downstream quality by converting OCR text into cleaner JSON using an LLM-based post-processing loop, reporting major BLEU improvement over the base model [54]. For your use case, this suggests a practical architecture: raw OCR -> structured extraction -> normalization into registry-ready schema [2] [54].

1.4 Cadastral maps and survey plans

A particularly relevant subfield for land systems is **map OCR plus spatial reasoning** [5] [6] [7]. Studies show that:

- detecting label orientation helps standard OCR on raster maps [6];
- OCRing latitude/longitude labels can automate georeferencing of scanned maps [5];
- spatial arrangement and neighboring context improve OCR correction on map text [7];
- prototype systems have processed actual land-register maps with arbitrary handwritten text positions and sizes [55].

For a verified land marketplace, this is strategically important because ownership validation often depends not just on names and dates, but also on **parcel labels, coordinates, and boundary alignment** [5] [7] [56].

Confidence level: Medium-High - the OCR/document-AI pattern is clear, but land-specific benchmark evidence remains limited [32] [33].

2. AI Validation and Fraud Detection for Property Documents

2.1 Best direct match: pre-storage authenticity validation

This is the section most directly aligned with your idea. The strongest evidence is a **2025 hybrid AI-blockchain framework for land records** that combines **OCR, NER, and anomaly detection** before records are stored, reporting **94.2% OCR accuracy, 90.9% NER F1, 93.4% validation accuracy, and 88.9% anomaly-detection F1** [8]. This is unusually close to your target architecture: scan or import records, validate them with AI, and only then allow secure downstream use [8].

Across the broader evidence, AI validation is framed as a way to **reduce fraudulent data entering immutable systems** [8]. This matters because blockchain or tamper-evident storage can protect records after entry, but does not guarantee the correctness of input data [8] [57].

2.2 Fraud signals and anomaly detection

The source set supports hybrid verification approaches rather than single-method designs [58]. OCR and ML pipelines are used to extract key fields, check structure, and detect irregularities or manipulations [59] [60] [61] [62] [63]. But there is also a major gap: the retrieved evidence does **not** include strong comparative property-document fraud studies testing OCR-only against OCR+NLP or multimodal verification for **stamps, seals, signatures, or altered dates** [36] [37] [38] [64].

So, the evidence supports **validation architecture**, but not yet a mature property-document fraud benchmark literature [8] [38].

2.3 Risk scoring and human review

The literature does not support full autonomy. Instead, it repeatedly recommends **AI-assisted screening with human oversight** [9] [10]. A municipal property-use workflow study reported **ROC-AUC 0.778, PR-AUC 0.720, F1 0.719, recall 0.887, precision 0.604**, while flagging **66.8% of cases** for prioritized human review [11]. That pattern is highly relevant to your proposed platform: use AI to create a **document trust score** and route uncertain cases to experts [11].

Confidence level: High - this is the strongest evidence cluster for your topic, especially the 2025 validation framework [8].

3. Record Linkage and Registry Integration

Validation of one scanned document is not enough if ownership truth is distributed across multiple government systems [12] [13]. The literature repeatedly points to fragmentation between **cadastre, land registry, address, survey, and tax systems** [12] [13] [65].

3.1 Why linkage matters

Record linkage research defines the problem as matching records from different datasets that refer to the same real-world entity [16] [66]. Foundational theory from Fellegi and Sunter formalized linkage as a **link / non-link / possible-link** decision problem [16]. Later work added EM estimation, assignment methods, blocking, and clerical review to make large-scale matching practical [29] [67] [68] [69].

For land systems, these methods are relevant to linking:

- owner names across registries,
- parcel identifiers across cadastral and transaction databases,
- addresses across municipal and tax systems,
- historical transaction chains across fragmented records [12] [13] [70].

3.2 Current land-specific interoperability evidence

The land-administration literature is stronger on **semantic and schema integration** than on published probabilistic owner-matching studies [14] [15] [17] [18]. Key contributions include:

- **LADM-based semantic models** for shared land-domain vocabulary [14] [71];
- **Linked data / knowledge organization systems** for standardization across land resources [17] [72];
- conceptual and object-relational integration of **land registry, cadastre, municipal, and tax data** [15] [18] [73] [74];
- integrated cadastral information systems linked to broader e-government and SDI needs [65] [75] [76].

The Spanish coordination studies show that cadastre-registry coordination often works through **spatial/parcel-boundary alignment**, not probabilistic record linkage [56] [77]. That is useful for your system because parcel geometry itself can become a validation signal [56].

3.3 Practical implication for your topic

Taken together, these findings suggest that a strong system should not act only as an OCR platform or listing portal. It should act as a **reconciliation layer** that checks whether document fields align with registry entries, cadastral parcels, survey geometry, and transaction history [12] [13] [15].

Confidence level: High - interoperability is a repeated, well-supported bottleneck across the land-administration evidence [12] [13] [14] [15].

4. Digital Land Administration and Trusted Property Transactions

4.1 eLAS and government land portals

Digital land administration systems are consistently described as improving **efficiency, transparency, accountability, and legal certainty** [19] [20] [21] [78]. More recent eLAS reviews also identify recurring barriers: **data integration gaps, stakeholder coordination, legal/regulatory weakness, incomplete digitization, and low digital literacy** [20] [25] [78].

Operational cases, including Malaysia and Indonesia-focused studies, show that online land services can streamline registration, validation, and public service delivery, but they remain constrained by **legacy paper systems, low data quality, public distrust, and uneven infrastructure** [19] [25] [41] [79] [80].

4.2 Blockchain and smart-contract transaction systems

Blockchain-based land systems are repeatedly proposed for **tamper resistance, immutability, auditability, and fraud reduction** [31] [81] [82] [83] [84] [85]. Smart contracts and permissioned ledgers can automate **verification, ownership transfer, and some dispute-handling logic** while reducing intermediaries and processing time [81] [82] [84] [86] [87] [88].

One especially strong technical result is a **Hyperledger Fabric** design with dynamic smart contracts that reduced average ownership transfer time to **420 ms from 5600 ms**, exceeded **175 TPS**, and was framed as suitable for legal compliance in land governance [24]. Other pilot-scale and rollout studies report **2-3 second transaction times** and major processing-time reductions [89] [90] [91].

4.3 Why validation must happen before immutable storage

A crucial finding across the literature is that blockchain does **not** solve the bad-input problem [8] [57]. One source explicitly states blockchain does not guarantee reliability of underlying information [57]. This is why your idea of **AI validation before listing or transaction use** is well aligned with the evidence [8].

4.4 Verified-document transaction platforms vs ad-driven marketplaces

The retrieved papers do not show a mature literature on full "intelligent property marketplaces" replacing advertisements entirely [22] [23] [26]. What they do show is a transition toward **verified transaction platforms**:

- blockchain-based systems using verified ownership records rather than ad-style listings [22] [23];
- portals with explicit verification steps such as seller/buyer workflows, e-stamp checks, and registry-office confirmation [26];
- systems that cross-reference external registries or oracles to confirm land details [28].

Inferred: these findings suggest that the most defensible marketplace model is not "property ads with AI," but **verification-backed transaction opportunities**, where visibility depends on document completeness, consistency, and legal freshness rather than marketing spend [8] [11] [28].
Confidence level: Low-Medium - this is a synthesis across transaction and validation studies, not a directly benchmarked marketplace result [8] [11] [28].

5. Best-Matching Research Paper Clusters for Your Work

If your goal is to collect the most relevant literature, these are the **best-matching clusters**:

Tier 1: Must-read, closest to your exact use case

- **AI validation before storage/use:** [8]
- **Trusted transaction platforms using verified ownership records:** [22] [23] [26] [28]

- **Interoperability and registry integration:** [12] [13] [14] [15]

Tier 2: Strong supporting methods

- **OCR preprocessing + structured extraction:** [1] [2] [3] [27]
- **Map OCR / georeferencing / cadastral text extraction:** [5] [6] [7] [55]
- **Risk scoring and human review prioritization:** [11]

Tier 3: Architecture and deployment context

- **eLAS / digital land administration:** [19] [20] [21] [25]
- **Permissioned blockchain / smart contracts for transfer:** [24] [89] [92] [93]

6. Recommended Research Architecture for Your Thesis or Product

A literature-backed system architecture would be:

Government records/scanned documents -> preprocessing -> OCR -> layout-aware extraction / NER -> rule-based validation -> anomaly detection -> cross-registry linkage -> human review queue -> audited storage / permissioned ledger -> verified-document transaction interface [1] [2] [8] [12].

Key design implications from the literature:

- do not trust OCR output directly [1] [2];
- validate before immutable storage [8] [57];
- treat cross-registry consistency as part of authenticity [12] [13];
- keep humans in the loop for uncertain or high-risk cases [9] [10] [69];
- design the marketplace around **trust scores and verified workflows**, not just listing exposure [11] [26].

Confidence level: High - this architecture is directly synthesized from the strongest recurring patterns in the evidence [1] [8] [12].

Conclusion

Yes—there is a **strong research match** for your topic, but it appears as a **combination of adjacent literatures**, not as one single mature field [1] [8] [12] [19]. The best-supported idea is a **verified-document-first land platform** in which OCR and document AI are only the front end; the real trust comes from **AI validation, anomaly detection, cross-registry linkage, human review, and auditable transaction workflows** [8] [13] [24].

The strongest single evidence item is the **2025 AI-blockchain land-record validation framework** [8]. The biggest research gaps are **property-specific OCR benchmarks**, **property-document fraud datasets**, and **buyer-facing intelligent marketplace studies** [22] [32] [36].

Overall confidence level: High - the system-level fit to your topic is strong, though some subcomponents, especially property-specific document AI benchmarking and marketplace design, remain underdeveloped in the available literature [8] [26] [33].

References

1. Jentsch, P., Porada, S. (2020) From Text to Data: Digitization, Text Analysis and Corpus Linguistics. *Digital Methods in the Humanities: Challenges, Ideas, Perspectives*. https://doi.org/10.14361/9783839454190-004
2. Gurbuz, A.S., Nassar, A., Auer, C., (...), Staar, P. (2027) Identify, Locate, Link: End-to-End Key-Value Extraction from Document Images. *Lecture Notes in Computer Science*. https://doi.org/10.1007/978-3-032-36033-5_22
3. Yoon, C.O., Lee, W., Jang, S., (...), Choi, D. (2024) Language, OCR, Form Independent (LOFI) pipeline for Industrial Document Information Extraction. *EMNLP 2024 - 2024 Conference on Empirical Methods in Natural Language Processing, Proceedings of the Industry Track*. https://doi.org/10.18653/v1/2024.emnlp-industry.79
4. Dereza, O., Rouget, M.-L., Egwu, C., Joy, C. (2025) Transforming Text into Discovery: OCR Enrichment of Digital Collections in the University of Galway Library. *Colloquium in Information Science and Technology, CIST*. https://doi.org/10.1109/CiSt65886.2025.11224240
5. Tyagi, P., Dubey, V. (2026) Automated Georeferencing of Topographic Maps via OCR and In-Context Multimodal LLM Reasoning. *Communications in Computer and Information Science*. https://doi.org/10.1007/978-3-032-08511-5_8
6. Chiang, Y.-Y., Knoblock, C.A. (2010) An approach for recognizing text labels in raster maps. *Proceedings - International Conference on Pattern Recognition*. https://doi.org/10.1109/ICPR.2010.783
7. Namgung, M., Chiang, Y.-Y. (2022) Incorporating spatial context for post-OCR in map images. *Proceedings of the 5th ACM SIGSPATIAL International Workshop on AI for Geographic Knowledge Discovery, GeoAI 2022*. https://doi.org/10.1145/3557918.3565864
8. Nagaraja, S.R., Shushma, R. (2025) A Hybrid AI-Blockchain Framework with OCR, NER, and Anomaly Detection for Secure Land Record Management. *Engineering, Technology and Applied Science Research*. https://doi.org/10.48084/etasr.15223
9. Dešić, J., Belanić, L. (2026) ARTIFICIAL INTELLIGENCE IN THE DIGITAL TRANSFORMATION OF LAND REGISTRATION PROCEDURES: BETWEEN EFFICIENCY AND LEGAL CERTAINTY. *Collected Papers of the Faculty of Law of the University of Rijeka*. https://doi.org/10.30925/zpfsr.47.1.16
10. Rybkina, A.M., Korobitsyna, E.S., Demidova, P.M., Shobairi, S.O.R. (2023) Artificial intelligence to assess the prospects of developing sites within the boundaries of roadsides and railway security zones. *AIP Conference Proceedings*. https://doi.org/10.1063/5.0132235

11. Francesconi, C., Valdebenito, R., Aguirre, C., Forcael, E. (2026) Screening Municipal Building Permissions for Sustainable Urban Renewal: An Interpretable Machine Learning Approach with Temporal Validation. *Buildings*. https://doi.org/10.3390/buildings16183667
12. Çağdaş, V., Stubkjaer, E. (2015) Core immovable property vocabulary for european linked land administration. *Survey Review*. https://doi.org/10.1179/1752270614Y.0000000093
13. Kalantari, M., Rajabifard, A., Wallace, J., Williamson, I. (2008) Spatially referenced legal property objects. *Land Use Policy*. https://doi.org/10.1016/j.landusepol.2007.04.004
14. Bošković, D., Ristić, A., Govedarica, M., Pržulj, D. (2010) Ontology development for land administration. *SIISY 2010 - 8th IEEE International Symposium on Intelligent Systems and Informatics*. https://doi.org/10.1109/SISY.2010.5647346
15. de Oliveira Farias, M., Carneiro, A.F.T., de Oliveira Lima, C. (2024) Integration of Geospatial and Alphanumeric data in the Structuring of a Database for Urban Registration. *Revista Brasileira de Cartografia*. https://doi.org/10.14393/rbcv76n0a-62960
16. Fellegi, I.P., Sunter, A.B. (1969) A Theory for Record Linkage. *Journal of the American Statistical Association*. https://doi.org/10.1080/01621459.1969.10501049
17. Stubkjaer, E., Whittal, J., Hull, S., (...), Folmer, E. (2025) The governance of global land sector semantic resources–status and pathways. *Survey Review*. https://doi.org/10.1080/00396265.2025.2543611
18. Dos Anjos Paiva, C.A., Antunes, A.F.B., Phillipi Camboim, S. (2018) A proposal for integrating data of land registry and urban Cadastre. *Boletim de Ciencias Geodesicas*. https://doi.org/10.1590/S1982-21702018000400032
19. Silviana, A., Elvani, R., Priyono, E.A. (2025) The Digitalization of Land Titles as an Effort to Achieve Good Land Governance in the Framework of Reduced Inequalities. *IOP Conference Series: Earth and Environmental Science*. https://doi.org/10.1088/1755-1315/1537/1/012031
20. Idris, K.A., Hamzah, A.H. (2025) The Conceptual Framework for Implementing Electronic Land Administration System Towards Sustainable Land Administration: Systematic Literature Review. *IEEE Symposium on Computer Applications and Industrial Electronics, ISCAIE*. https://doi.org/10.1109/ISCAIE64985.2025.11080960
21. Prasetya, F., Mahfud, M. (2023) Pendaftaran Tanah Untuk Pertama Kali Secara Elektronik Dalam Hukum Pertanahan Nasional. *Jurnal Hukum Unissula*. https://doi.org/10.26532/jh.v39i1.30581
22. https://www.sciencedirect.com/science/article/pii/S157375432600151X
23. Engin, Zeynep, Treleaven, Philip (2018) Algorithmic Government: Automating Public Services and Supporting Civil Servants in using Data Science Technologies. *Computer Journal*. https://doi.org/10.1093/comjnl/bxy082

24. Chandra, S., Elrashidi, A., Purushotham, S., (...), Dinesh, M. (2025) A Blockchain based Secured Land Registration System using Dynamic Smart Contracts. *Proceedings - 2025 International Conference on Recent Innovation in Science Engineering and Technology, ICRISET 2025*. https://doi.org/10.1109/ICRISET64803.2025.11252328

25. Idris, K.A., Hamzah, A.H. (2025) Electronic Land Administration System for Sustainable Development: Case Study of e-Tanah System Implementation. *IEEE Symposium on Computer Applications and Industrial Electronics, ISCAIE*. https://doi.org/10.1109/ISCAIE64985.2025.11081072

26. https://www.sciencedirect.com/science/article/pii/S1936645021001625

27. Viswapriya, S.E., Preethi, R., Rajalakshmi, R. (2025) AI-Based Tools for Legal Data Extraction using Deep Learning Techniques. *2025 IEEE International Conference on Advanced Healthcare Systems: Architecture, Engineering and Management, ICAHS 2025*. https://doi.org/10.1109/ICAHS66950.2025.11553091

28. Habib, Md. Ahsan, Manik, Md. Motaleb Hossen (2026) Blockchain-enabled secure land record management system with a new lightweight cryptosystem based on a hybrid chaotic map. *Blockchain: Research and Applications*. https://doi.org/10.1016/j.bcra.2026.100531

29. Jaro, M.A. (1989) Advances in record-linkage methodology as applied to matching the 1985 census of Tampa, Florida. *Journal of the American Statistical Association*. https://doi.org/10.1080/01621459.1989.10478785

30. Herzog, T.N., Scheuren, F.J., Winkler, W.E. (2007) Data quality and record linkage techniques. *Data Quality and Record Linkage Techniques*. https://doi.org/10.1007/0-387-69505-2

31. Gedik, Ü., Ercan, O. (2026) Assessing Blockchain Integration in the Turkish Land Registry System: A Conceptual Model. *Turkish Journal of Engineering*. https://doi.org/10.31127/tuje.1895953

32. Zhang, J., Zhang, Y. (2025) DocRouter: Prompt guided vision transformer and Mixture of Experts connector for document understanding. *Information Fusion*. https://doi.org/10.1016/j.inffus.2025.103206

33. Liu, S., Zhang, Z., Hu, P., (...), Liu, C. (2026) See then tell: Enhancing key information extraction with vision grounding. *Neurocomputing*. https://doi.org/10.1016/j.neucom.2026.132858

34. Vermote, E., Justice, C., Claverie, M., Franch Gras, B. (2016) Preliminary analysis of the performance of the Landsat 8/OLI land surface reflectance product. *Remote Sensing of Environment*. https://doi.org/10.1016/j.rse.2016.04.008

35. Roy, D.P., Zhang, H.K., Ju, J., (...), Kovalskyy, V. (2016) A general method to normalize Landsat reflectance data to nadir BRDF adjusted reflectance. *Remote Sensing of Environment*. https://doi.org/10.1016/j.rse.2016.01.023

36. https://www.sciencedirect.com/science/article/pii/S2662998426006987

37. Shahriar, S., Dara, R., Lund, B., Kamali, M. (2027) A comparative study on textual and multimodal PII detection using large language models. *Computer Standards and Interfaces*. https://doi.org/10.1016/j.csi.2026.104185

38. https://www.sciencedirect.com/science/article/pii/S1573772124020088

39. Feng, Chen, Tao, Yunqing, Zhang, Yan, Zhu, Xingshu (2024) Local land regulatory governance and land transaction prices: Micro evidence from land audits. *Economic Analysis and Policy*. https://doi.org/10.1016/j.eap.2024.08.011

40. Rashid, M.R.A., Al-Rafi, A.A., Islam, M.A., (...), Khan, M.S.H. (2025) Enhancing land management policy in Bangladesh: A blockchain-based framework for transparent and efficient land management. *Land Use Policy*. https://doi.org/10.1016/j.landusepol.2024.107436

41. Jahani Chehrebargh, F., Rajabifard, A., Atazadeh, B., Steudler, D. (2024) Current challenges and strategic directions for land administration system modernisation in Indonesia. *Journal of Spatial Science*. https://doi.org/10.1080/14498596.2024.2360531

42. Rybalkin, V., Ghafoor, A., Bukhari, S.S., (...), Dengel, A. (2018) Idocchip - A configurable hardware architecture for historical document image processing: Percentile based binarization. *Proceedings of the ACM Symposium on Document Engineering 2018, DocEng 2018*. https://doi.org/10.1145/3209280.3209538

43. Jadhav, P., Mayuree, M., Anushka, A., (...), Deshpande, P. (2022) Pix2Pix Generative Adversarial Network with ResNet for Document Image Denoising. *4th International Conference on Inventive Research in Computing Applications, ICIRCA 2022 - Proceedings*. https://doi.org/10.1109/ICIRCA54612.2022.9985695

44. Sai Rakesh Kamisetty, V.N., Sohan Chidvilas, B., Revathy, S., (...), Gladence, L. (2022) Digitization of Data from Invoice using OCR. *Proceedings - 6th International Conference on Computing Methodologies and Communication, ICCMC 2022*. https://doi.org/10.1109/ICCMC53470.2022.9754117

45. Akoushideh, A., Ranjkesh Rashtehroudi, A., Shahbahrami, A. (2025) Persian/Arabic Scene Text Recognition With Convolutional Recurrent Neural Network. *IET Smart Cities*. https://doi.org/10.1049/smc2.70001

46. Bukhari, S.S., Francis, S., Kamath, C.N.N., Dengel, A. (2018) An investigative analysis of different LSTM libraries for supervised and unsupervised architectures of OCR training. *Proceedings of International Conference on Frontiers in Handwriting Recognition, ICFHR*. https://doi.org/10.1109/ICFHR-2018.2018.00084

47. Chandra, A., Stefanus, R. (2021) An End-to-End Optical Character Recognition Pipeline for Indonesian Identity Card. *2021 9th International Conference on Information and Communication Technology, ICoICT 2021*. https://doi.org/10.1109/ICoICT52021.2021.9527436

48. Yindumathi, K.M., Chaudhari, S.S., Ramalingaiah, R. (2020) Analysis of Image Classification for Text Extraction from Bills and Invoices. *2020 11th International Conference on Computing, Communication and Networking Technologies, ICCCNT 2020*. https://doi.org/10.1109/ICCCNT49239.2020.9225564

49. Naqi Hadi, M.A., Gul, M., Khan, M., (...), Jhanjhi, N. (2024) Benchmarking Performance Analysis of Optical Character Recognition Techniques. *Proceedings of the IEEE International Multi Topic Conference, INMIC*. https://doi.org/10.1109/INMIC64792.2024.11004392

50. Novotný, V., Horák, A. (2022) When Tesseract Meets PERO Open-Source Optical Character Recognition of Medieval Texts. *Recent Advances in Slavonic Natural Language Processing*. https://www.scopus.com/pages/publications/85163814694
51. de Rodrigo, I., López-López, A.J., Boal, J. (2026) VERSE: Visual Embedding Reduction and Space Exploration — Latent-space clustering for improving document understanding. *Pattern Recognition*. https://doi.org/10.1016/j.patcog.2026.114448
52. de Rodrigo, I., Sanchez-Cuadrado, A., Boal, J., López-López, A.J. (2026) The MERIT dataset: Modelling and efficiently rendering interpretable transcripts. *Pattern Recognition*. https://doi.org/10.1016/j.patcog.2025.112502
53. Billah, M.M., Ahmed, M.U., Barua, S., (...), Begum, S. (2026) Evaluating AI Agent prompting and fine-tuning strategies for structured data extraction from industrial PDFs. *Array*. https://doi.org/10.1016/j.array.2026.101118
54. Abdellatif, O.H., Ayman, A., Hamdi, A. (2025) LLMOps-Driven Robotic Process Automation Approach. *3rd International Conference on Intelligent Methods, Systems and Applications, IMSA 2025*. https://doi.org/10.1109/IMSA65733.2025.11167061
55. Di Zenzo, S., del Buono, M., Meucci, M., Spirito, A. (1992) Optical recognition of hand-printed characters of any size, position, and orientation. *IBM Journal of Research and Development*. https://doi.org/10.1147/rd.363.0487
56. Mora-Navarro, G., Ribera, C., Velilla-Torres, J.M., Martinez-Llario, J. (2022) Geographical Data and Metadata on Land Administration in Spain. *Land*. https://doi.org/10.3390/land11071107
57. Lemieux, V.L. (2016) Trusting records: is Blockchain technology the answer?. *Records Management Journal*. https://doi.org/10.1108/RMJ-12-2015-0042
58. Uppar, A., Veerakyatharayappa, N., Chikkanna, P. (2026) Cybersecurity in property digitisation: a taxonomy and evaluation of current security methods. *Bulletin of Electrical Engineering and Informatics*. https://doi.org/10.11591/eei.v15i3.10945
59. Dhutraj, N., Sonawane, S., Patil, M., (...), Patil, J. (2026) Automated Document Verification for Official Documents. *17th International Conference on Advances in Computing, Control, and Telecommunication Technologies, ACT 2026*. https://www.scopus.com/pages/publications/105047257047
60. Dai, K., Alonso, J., Gutiérrez-Meana, J. (2025) A Machine Learning Framework for Forgery Detection in Digital ID Documents. *Journal of Innovation Management*. https://doi.org/10.24840/2183-0606_013.002_L003
61. Raut, A., Puri, C. (2026) XAI-Based Document Verification System for Government Services and Job Applications. *7th International Conference on Mobile Computing and Sustainable Informatics, ICMCSI 2026*. https://doi.org/10.1109/ICMCSI67283.2026.11412546

62. Tutaya, R.G.Q., Gaspar, G.Y.A., Franco, C.J.R. (2025) Application of a Technological Model for the Automated Verification of Vehicle Documents through Computer Vision in the City of Huancayo. *ICTLE 2025 - 2025 13th International Conference on Traffic and Logistic Engineering*. https://doi.org/10.1109/ICTLE67020.2025.11203620

63. He, S. (2025) Efficiency and accuracy improvement of document verification: Automation and method analysis. *Proceedings of SPIE - The International Society for Optical Engineering*. https://doi.org/10.1117/12.3067142

64. Luo, Qiling, Cheng, Boyuan, Li, Mingchao, (...), Zhang, Jinrui (2025) Real-world mega-project data and particle-packing features drive AI design of marine high-performance concrete. *Case Studies in Construction Materials*. https://doi.org/10.1016/j.cscm.2025.e05306

65. Krigsholm, P., Riekkinen, K., Ståhle, P. (2018) The changing uses of cadastral information: A user-driven case study. *Land*. https://doi.org/10.3390/land7030083

66. Christen, P., Gayler, R., Hawking, D. (2009) Similarity-aware indexing for real-time entity resolution. *International Conference on Information and Knowledge Management, Proceedings*. https://doi.org/10.1145/1645953.1646173

67. McCallum, A., Nigam, K., Ungar, L.H. (2000) Efficient clustering of high-dimensional data sets with application to reference matching. *Proceeding of the Sixth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. https://doi.org/10.1145/347090.347123

68. Jaro, M.A. (1995) Probabilistic linkage of large public health data files. *Statistics in Medicine*. https://doi.org/10.1002/sim.4780140510

69. Larsen, M.D., Rubin, D.B. (2001) Iterative automated record linkage using mixture models. *Journal of the American Statistical Association*. https://doi.org/10.1198/016214501750332956

70. Champney, T.F., Qin, H. (2026) Maximizing Linkage in Address Data: Spatial, Exact, and Fuzzy Matching. *Journal of Data Science*. https://doi.org/10.6339/26-JDS1236

71. Sladić, D., Radulović, A., Govedarica, M., (...), Pržulj, Đ. (2015) The use of ontologies in cadastral systems. *Computer Science and Information Systems*. https://doi.org/10.2298/CSIS141031009S

72. Beek, W., Folmer, E. (2017) An integrated approach for linked data browsing. *International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences - ISPRS Archives*. https://doi.org/10.5194/isprs-archives-XLII-4-W2-35-2017

73. Giordano, D., Torre, A., Samperi, C., (...), Faro, A. (2013) An ontology based approach to integrate data and maps in the government enterprise architecture: A case study. *ICEIS 2013 - Proceedings of the 15th International Conference on Enterprise Information Systems*. https://www.scopus.com/pages/publications/84887709354

74. Giordano, D., Torre, A., Samperi, C., (...), Faro, A. (2014) Joining data and maps in the government enterprise architecture by a semantic approach: Methodology, ontology and case study. *Lecture Notes in Business Information Processing*. https://doi.org/10.1007/978-3-319-09492-2_30

75. Vahala, M. (2004) Towards integrated work processes: Finnish Land Registry and Cadastre development strategy. *GIM International*. https://www.scopus.com/pages/publications/1642359221

76. Rajabifard, A., Williamson, I., Steudler, D., (...), King, M. (2007) Assessing the worldwide comparison of cadastral systems. *Land Use Policy*. https://doi.org/10.1016/j.landusepol.2005.11.005

77. Ribera, C., Mora-Navarro, G., Martinez-Llario, J.C. (2021) Advances in the coordination between the cadastre and land registry. *Land*. https://doi.org/10.3390/land10010081

78. Idris, K.A. (2024) The Implementation an Electronic Land Administration System Towards Sustainable Land Administration: Systematic Literature Review. *14th IEEE Symposium on Computer Applications and Industrial Electronics, ISCAIE 2024*. https://doi.org/10.1109/ISCAIE61308.2024.10576247

79. Bennett, R.M., Hutabarat, I.R., Unger, E.-M., (...), Latif, A. (2023) Digital Transformation in Indonesian Land Administration: Further Developments and Directions. *Geospatial Science for Smart Land Management: An Asian Context*. https://doi.org/10.1201/9781003349518-7

80. Akter, M. (2022) Digitalization in the Land Service Delivery: Comparison between Bangladesh and Indonesia. *Southeast Asia: A Multidisciplinary Journal*. https://doi.org/10.1108/SEAMJ-01-2022-B1006

81. Junaid, L., Bilal, K., Shuja, J., (...), Rodrigues, J.J.P.C. (2024) Blockchain-Enabled Framework for Transparent Land Lease and Mortgage Management. *IEEE Access*. https://doi.org/10.1109/ACCESS.2024.3388248

82. Ramana Lakshmi, A., Nandini, C., Rayudu, G.V.M.D., (...), Sai, B.J. (2024) Digital Land Registration Using Smart Contracts. *Proceedings - 2nd International Conference on Advancement in Computation and Computer Technologies, InCACCT 2024*. https://doi.org/10.1109/InCACCT61598.2024.10551084

83. Vinothiyalakshmi, P., Muralidharan, C., Anitha, R., Yoosuf, Y.M. (2022) Digitized Land Registration Using Blockchain Technology. *Blockchain Technology: Exploring Opportunities, Challenges, and Applications*. https://doi.org/10.1201/9781003138082-5

84. Avula, A., Sammeta, N. (2026) LandReg: A Multi-Signature NFT-Based Blockchain Land Registry and Ownership Verification System. *2026 International Conference on Emerging Trends in Mobile Computing and Sustainable Informatics, ICEMCSI 2026*. https://doi.org/10.1109/ICEMCSI67638.2026.11603219

85. Mansoor, M.A., Ali, M., Mateen, A., (...), Nazir, S. (2023) Blockchain Technology for Land Registry Management in Developing Countries. *2023 2nd International Conference on Emerging Trends in Electrical, Control, and Telecommunication Engineering, ETECTE 2023 - Proceedings*. https://doi.org/10.1109/ETECTE59617.2023.10396736

86. Swetha, M.S., Yadav, S.A., Rao, S., (...), Pathak, N. (2025) Blockchain Technology to Modernize the Process of Land Registration and Ownership Hierarchy. *Lecture Notes in Networks and Systems*. https://doi.org/10.1007/978-981-96-6124-4_49

87. Borse, Y., Ahirao, P., Chawathe, A., Patole, D. (2022) Embracing Permissioned Blockchain for a Secure and Reliable Land Registration System. *5th IEEE International Conference on Advances in Science and Technology, ICAST 2022*. https://doi.org/10.1109/ICAST55766.2022.10039485

88. Mallya, A., Ananth, A., Baliga, A.R., (...), Honnavalli, P.B. (2025) SLR-GPoA: A Smart Land Registry System Using Blockchain and Geographic Proof of Authority. *Journal of Advances in Information Technology*. https://doi.org/10.12720/jait.16.12.1758-1779

89. Gadde, Y., Jyothi, B.P., Murari, C., Vijayaraj, N. (2026) Tokenized land ownership: A decentralized blockchain-based land record system using NFTs. *Artificial Intelligence, Computational Intelligence, and Inclusive Technologies: Proceedings of International Conference on Artificial Intelligence, Computational Intelligence, and Inclusive Technologies (ICRAIC2IT – 2025)*. https://doi.org/10.1201/9781003740100-103

90. Dhanalakshmi, B.K., Lakshmi, B.N., Archana, R.A., Udayaprasad, P.K. (2026) Blockchain-Based Decentralized Framework for Secure and Transparent Land Registration: Enhancing Data Integrity and Reducing Intermediaries. *Lecture Notes in Networks and Systems*. https://doi.org/10.1007/978-981-95-3492-0_12

91. Maheswari, C., Lavanya, K., Harini, G., Kumar, A.B. (2025) Efficient Approach for Land Registration using Blockchain Technology. *16th International Conference on Advances in Computing, Control, and Telecommunication Technologies, ACT 2025*. https://www.scopus.com/pages/publications/105034857821

92. Albataineh, A.F. (2025) The effect of smart contracts (blockchain technology) on Jordan's land registry and survey department for sustainable development. *Discover Sustainability*. https://doi.org/10.1007/s43621-025-00797-7

93. von Wangenheim, G. (2020) Blockchain-based land registers: A law-and-economics perspective. *Disruptive Technology, Legal Innovation, and the Future of Real Estate*. https://doi.org/10.1007/978-3-030-52387-9_6